Validation of the kinematic and dynamic accuracy of a depth-camera-based telerehabilitation system

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Introduction

Markerless motion capture systems for telerehabilitation that enable quantitative monitoring of physical activity are becoming a reality. Our group developed Muvity (https://muvity.cat/; see the feasibility study in [1]), a telerehabilitation platform designed for people with frailty. Muvity allows users to perform therapeutic exercises and exergames while interacting with a software application through body movements captured by a depth camera. The system estimates upper-limb kinematics and dynamics, including joint moments and powers, and reports these metrics to the therapist.

In this study, we evaluated the accuracy of Muvity by comparing the upper-limb kinematics and dynamics estimated by the system with those obtained using an optical motion capture system. The aim was to validate the accuracy of the biomechanical data reported to therapists.

Methods

A total of 10 healthy participants (26 ± 7 years, 67 ± 12 kg) were assessed in the motion capture laboratory at EEBE-UPC. Three data acquisition systems were used simultaneously: an optical motion capture system (OptiTrack Flex 13) to record the trajectories of 40 skin-mounted reflective markers; the Muvity system, using an Orbbec Gemini depth camera, to estimate body kinematics; and two AMTI AccuGait force platforms to record ground reaction forces. The three systems were temporally synchronized, and their coordinate systems were spatially aligned. Ethical approval was obtained from the Ethics Committee of the Universitat Politècnica de Catalunya (CEUPC, reference 2024-011).

Participants performed nine movements involving the upper and lower limbs and covering the main human degrees of freedom (DoFs) of upper and lower limbs. Following data processing, the kinematic and kinetic outputs obtained with the two motion analysis systems were compared. Marker trajectories recorded by the optical system were processed using inverse kinematics in OpenSim [2] to estimate joint coordinates. Inverse dynamics was subsequently applied to estimate joint moments, incorporating the measured ground reaction forces for lower-limb movements. Joint powers were then calculated from the resulting joint moments and angular velocities. Muvity estimates joint coordinates internally and performs inverse dynamics using the OpenSim application programming interface.

Results

Joint angles for the flexion-extension and abduction-adduction DOFs showed RMSE values ranging from 5° to 15°. For example, for one subject performing a pure shoulder flexion movement, the shoulder flexion and adduction angles had RMSE values of 5.9° and 7.6°, respectively (Figure 1). Shoulder joint moments showed RMSE values below 4 Nm, with high correlation coefficients (r = 0.97 and 0.82 for the flexion-extension and abduction-adduction moments, respectively). For the same subject and movement, the corresponding RMSE values for shoulder joint power were 10.6 W and 0.6 W, with correlation coefficients of r = 0.99 and 0.86, respectively.

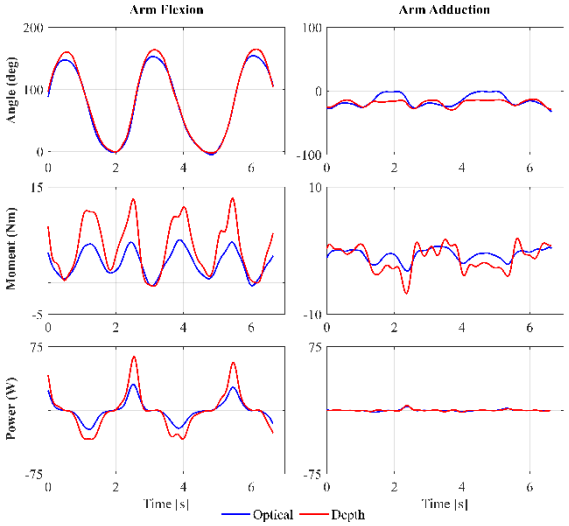


Figure 1. Example of shoulder joint angles, moments and powers for the flexion-extension, and abduction-adduction DoFs during a pure shoulder flexion movement.

Conclusions

This study demonstrates that a markerless telerehabilitation system can accurately estimate upper-limb kinematics and dynamics of the main DoFs, supporting its use for objective remote functional assessment.

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References

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